

NAMAN KANDPAL

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EXPERIENCE

Astroverse Experience Private Limited

Haldwani, India

R&D Intern

July–September 2025

- Engineered and flight-tested a **full-stack IoT satellite**, validating a stable telemetry and video link over a **105-meter apogee** while streaming live data from a multi-sensor suite.
- Architected a **real-time, dual-core embedded system** on FreeRTOS, implementing **mutexes and ISR-driven semaphores** for concurrent-safe I2C communication and efficient IMU data processing.
- Re-engineered an **asynchronous ESP32 web server** and interactive dashboard, leveraging **WebSockets** for low-latency telemetry streaming and enabling client-side flight data logging to a downloadable CSV.

EDUCATION

Graphic Era Hill University

Bhimtal, India

Bachelor of Technology, Computer Science & Engineering

July 2022 – 2026

- Cumulative GPA: 8.28 / 10.0

White Hall School

Haldwani, India

Senior School Certificate Examination

July 2022

- Overall Percentage: 93%

Secondary School Certificate Examination

July 2020

- Overall Percentage: 92%

PROJECTS

Travelling Salesman Problem Testing Framework | *Python, Pandas, Matplotlib* 🐙 Github *March 2026*

- Developed an automated comparative evaluation suite to benchmark **Nearest Neighbour**, **2-Opt**, and **Genetic Algorithms** on complex topographical datasets scaling up to 2,500 cities.
- Authored a formal research paper applying these algorithmic solutions to real-world logistical challenges, specifically targeting efficient supply chain and tourism routing in the mountainous terrains of Uttarakhand.

Rubik's Cube Solver | *C++, OpenGL, Graph Algorithms* 🐙 Github

June 2025

- Developed a solver using **A* search** and **Thistlethwaite's Algorithm** to find optimal move sequences from any cube configuration in under 40 moves.
- Engineered a high-performance solution capable of analyzing several unique cube states per second, applying mathematical concepts like God's Number and group theory.

ACHIEVEMENTS

- Qualified GATE Computer Science 2026 Examination with **AIR 6123** (97th percentile).
- First Runner-up**, Inter-University Ethics Bowl (April 2025): Led the university team to 2nd place, demonstrating analytical thinking and ethical reasoning.
- Finalist**, GraByte Tech Fest (2024): Ranked in the top 5% (top 10 of 200+ participants) in a university-level competitive programming contest.

TECHNICAL SKILLS

Languages: C++, Python, C

Frameworks & Libraries: Matplotlib, NumPy, Pandas, FreeRTOS, Arduino

Developer Tools: Git, GitHub, VS Code, Github Copilot